

TEAM I
IT ARCHITECTS

CREDIT CARD FRAUD DETECTION



Agenda

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- Improvement from Professor
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- Team Working Agreement
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- Product Backlog
- Sprint 2 Stories
- Sprint 2 Test Cases
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- Sprint 2 Velocity
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- Committed and Completed Ratios
- Sprint Retrospective
- Sprint 3 Planning Stories Carried Over
- Application Screen Shot
- Rest API
- Wiki Page & GitHub Repository
- Project Demo

Team Members



Mr. Arth Patel

Role : Front-End Developer



Mr. Tyler Helmrich

Role: Scrum Master



Mr. Jay Patel

Role: Back-End Developer



Mr. Xavier Jackson

Role: Machine Learning Lead



Mr. Zenil Moradia

Role: Quality assurance

Improvement from Professor

- Some diagrams were incomplete or not done properly
- Missing required Agile documents:
 - Team Working Agreement
 - MVP definition
 - Product Backlog
 - Sprint Velocity tracking
- User stories were unclear and inconsistent.
- Overall documentation quality needed improvement

Problem Statement

Credit card fraud remains the leading form of identity theft in the United States, with 2024 seeing over 449,000 complaints to the Federal Trade Commission and within the first quarter of 2025 adding another 151,000 cases. In the United States, 63% of cardholders have been targeted at least once, and more than half of these victims report experiencing repeated attacks.

This persistent threat not only inflicts hundreds of millions of dollars in annual losses but also erodes consumer confidence in digital payments. Existing detection systems struggle to keep up with the evolving tactics of scammers and the extreme imbalance between legitimate and fraudulent transactions. There is an urgent need for a robust, real-time solution that can accurately flag while minimizing false alarms.

Problem Description

Project Name:	"Classifi" - Credit Card Fraud Detection Mobile App utilizing a Classification Algorithm Machine Learning Model
Team:	IT Architects
Project Description:	This project aims to deliver an advanced machine-learning classifier for credit-card fraud detection, designed for issuers and cardholders frustrated by today's rule-based systems and high false-alarm rates. Leveraging SMOTE to rebalance data and XGBoost for precision, we'll identify fraudulent transactions before funds leave the account, reduce false positives on legitimate purchases, and trigger real-time alerts that let cardholders act immediately. Our mission is to replace rigid, hand-tuned rules with a data-driven model that adapts to evolving fraud patterns by minimizing losses, preserving reputations, and restoring trust in digital payments through a mobile iOS application that is easily accessible. Current deterrents include rule-based fraud detection techniques, whereas our goal is to implement an AI-powered fraud-detection service that catches up to 80% of fraud attempts in real time while cutting false positives by 30%, so issuers lose less money and cardholders enjoy seamless, secure spending.
Benefit Outcomes:	Credit Card companies can expect a reduction in false alarms, monetary losses, and damage to their reputation. Credit Card holders can expect a reduction in unauthorized charges, fewer declined legitimate transactions, and greater peace of mind when using their cards as well as an increase in user satisfaction and engagement.
Github Link:	https://github.com/htmw/F2025-Team1/wiki

Team Working Agreement

- **Communication & Collaboration** - Our team will primarily communicate through **WhatsApp, Teams, Discord** and **GitHub** updates.
- Whenever code is pushed to GitHub, members will notify the team to prevent conflicts or overwrites.
- If anyone needs to adjust their task timeline or change their approach, they must inform the team beforehand.
- **Meeting Routine & Workflow** - The team will meet twice weekly, typically on Saturday and Wednesday, unless additional sessions are required.
- Anyone unable to attend a meeting must notify the team in advance and go through the meeting summary afterward.

Team Working Agreement Cont.

- **Team Working Principles** - All tasks will include a clear deadline and will be reviewed once completed.
- All contributions should be pushed to GitHub regularly with meaningful commit messages.
- Complex tasks will be broken into smaller actionable items to balance workload.

- **Checklist for Marking a Task as Complete** - The output should satisfy all acceptance criteria, such as correct preprocessing, expected model accuracy, or UI functionality.
- All associated documentation, dataset changes should be updated.

Persona 1: Sarah Miller



- Age: 28
- Occupation: Graphic Designer

- Sarah is an avid online shopper who loves finding unique items from small businesses. She uses her credit card for most of her purchases, appreciating the convenience and rewards. She's tech-savvy and expects seamless digital experiences. Recently, her card details were stolen from a third-party website, leading to several unauthorized small purchases. The process of getting a new card and updating her payment information on various subscriptions was a significant hassle and made her wary of future online transactions. She's looking for reassurance and security in her digital financial interactions.
- Goals:
 - Wants to shop online without constant worry about card fraud.
 - Needs quick and clear notifications for any suspicious activity on her accounts.
 - Hopes for an easy way to verify transactions or freeze her card directly from her phone if needed.
- Frustrations:
 - The time and effort involved in cancelling cards and updating payment details after fraud.
 - Feeling vulnerable and exposed when her financial information is compromised.
 - Receiving generic, unhelpful alerts that don't provide clear next steps.
- Needs from the Solution:
 - Real-time alerts that are easy to understand and act upon.
 - An intuitive mobile interface that allows her to quickly approve or deny a transaction and temporarily lock her card.
 - Minimal false positives on her legitimate purchases so she doesn't lose trust in the system.

Persona 2: Alex Chen



- Age: 32
- Occupation: Founder of "PetPals Boutique" | The E-commerce Startup Founder

- Age: 32
- Occupation: Founder of "PetPals Boutique" | The E-commerce Startup Founder
- Alex recently launched "PetPals Boutique," an online store selling custom pet accessories. Business is booming, but he's starting to see a disturbing trend of chargebacks due to fraudulent orders. These chargebacks not only cost him money but also damage his store's reputation with payment processors. He's passionate about his business but has limited resources and can't afford a dedicated fraud team. He needs automated, intelligent system that can protect his revenue and allow him to focus on growing PetPals Boutique, not fighting fraud.
- Goals:
 - To minimize chargebacks and financial losses from fraudulent transactions.
 - To protect his business's reputation with payment providers.
 - To have a simple, affordable, and effective fraud detection solution that integrates easily with his e-commerce platform.
- Frustrations:
 - The financial impact of fraudulent orders and chargeback fees.
 - The time-consuming process of manually reviewing suspicious orders.
 - Fear that his small business could be targeted by organized fraud rings, which he feels ill-equipped to handle.
- Needs from the Solution:
 - An easy-to-integrate API that works seamlessly with his e-commerce platform.
 - A dashboard that provides clear, actionable insights on potential fraud, without requiring a lot of technical expertise.
 - Automated decision-making that blocks high-risk transactions instantly and allows low-risk ones to proceed without delay.

Persona 3: Marcus Johnson



- Age: 49
- Occupation: Head of Fraud Prevention at "Secure Bank"

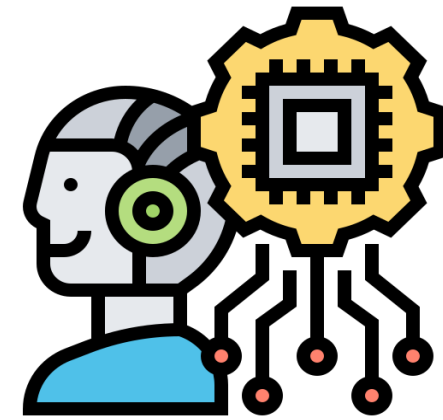
- Marcus leads the fraud prevention department at a mid-sized bank. His team handles thousands of credit card transactions daily, constantly battling sophisticated fraud schemes. With new tactics emerging all the time, his biggest challenge is staying ahead of the criminals while ensuring legitimate customer transactions are processed smoothly. He's looking for a powerful, scalable solution that can integrate with their existing systems, reduce false positives, and provide actionable intelligence to his team. He needs to improve detection rates, reduce operational costs associated with fraud investigation, and enhance customer trust.
- Goals:
 - To significantly reduce the bank's annual losses from credit card fraud.
 - To decrease the rate of false positives, improving customer experience and reducing operational overhead.
 - To implement a system that provides real-time, adaptive fraud detection capable of learning new patterns.
- Frustrations:
 - The constant arms race with fraudsters and the difficulty of keeping up with evolving tactics.
 - The pressure to maintain a high level of security without inconveniencing legitimate customers.
 - The sheer volume of data and the challenge of sifting through it to find genuine fraud.
- Needs from the Solution:
 - A highly scalable system that can process millions of transactions in real time.
 - Advanced machine learning capabilities that can adapt to new fraud patterns and minimize false positives.

MVP

- **Real-Time Fraud Predictions** – System analyzes incoming transactions instantly.
- **Interactive Dashboard** – Users can visualize fraud alerts, risk scores, and patterns through graphs & metrics.
- **Simple, Accessible UI** – Easy for non-technical users to understand fraud insights at a glance.
- **API-Driven Architecture** – Front-end fetches predictions seamlessly through backend REST APIs.
- **Notification/Alert System (TBD)** – Email/SMS alerts for high-risk transactions (future enhancement).



Technology and Tools



Data Storage, Collection, and Processing

Kaggle – Online repository of public datasets. We used this to acquire the training data set for our model.

Pandas / NumPy – Core tabular and numerical routines.

Scikit-learn – Standardized pipeline, preprocessing, and model-selection tools.

Supabase – an open-source backend-as-a-service (BaaS) platform that provided us with a full suite of tools—database, authentication, storage, real-time subscriptions, and serverless functions.



Model Development and Experimentation

XGBoost – Gradient boosting implementations with GPU support for speed and advanced class-imbalance handling.

Python – widely used language for AI model development because of its simplicity, readability, and powerful ecosystem of libraries and frameworks.

FastAPI – a modern, high-performance Python web framework designed for building APIs quickly and efficiently.



Visualization and End-User Interface

PenPot – a free, open-source, web-based design and prototyping tool built for collaboration between designers and developers.

XCODE – Apple’s official IDE for building apps across all Apple platforms.

Swift – Apple’s modern, powerful, and intuitive programming language designed for building Apple Platform apps on XCODE.

Algorithms and Models

Problem Solver

SMOTE

Synthetic Minority Over-sampling Technique.

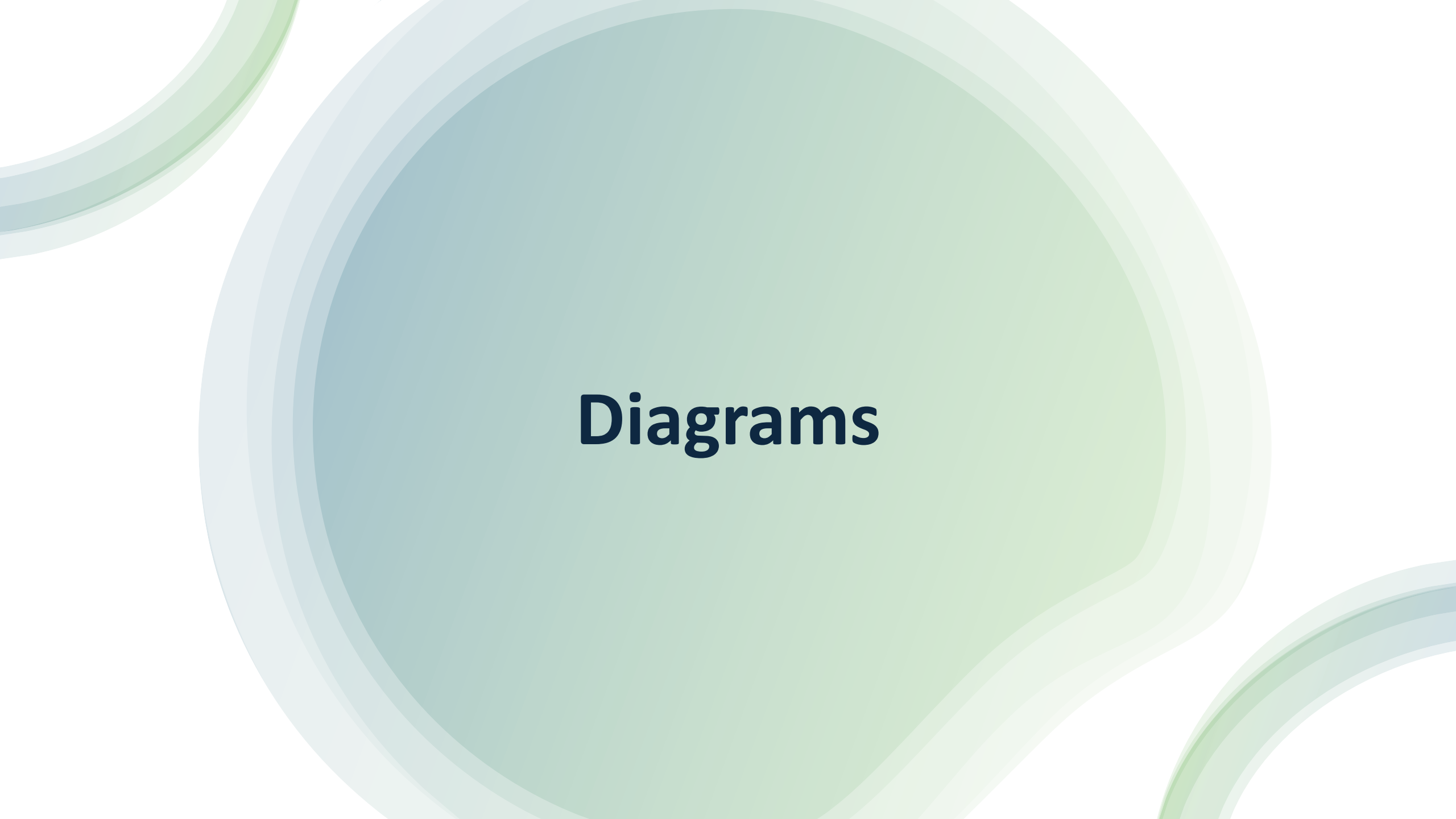
Used to mathematically generate new fraud examples on the **training data** to address the severe **class imbalance** (99.83% legitimate) and ensure the model learns fraud patterns.

Classifier

XGBoost

The **eXtreme Gradient Boosting** algorithm.

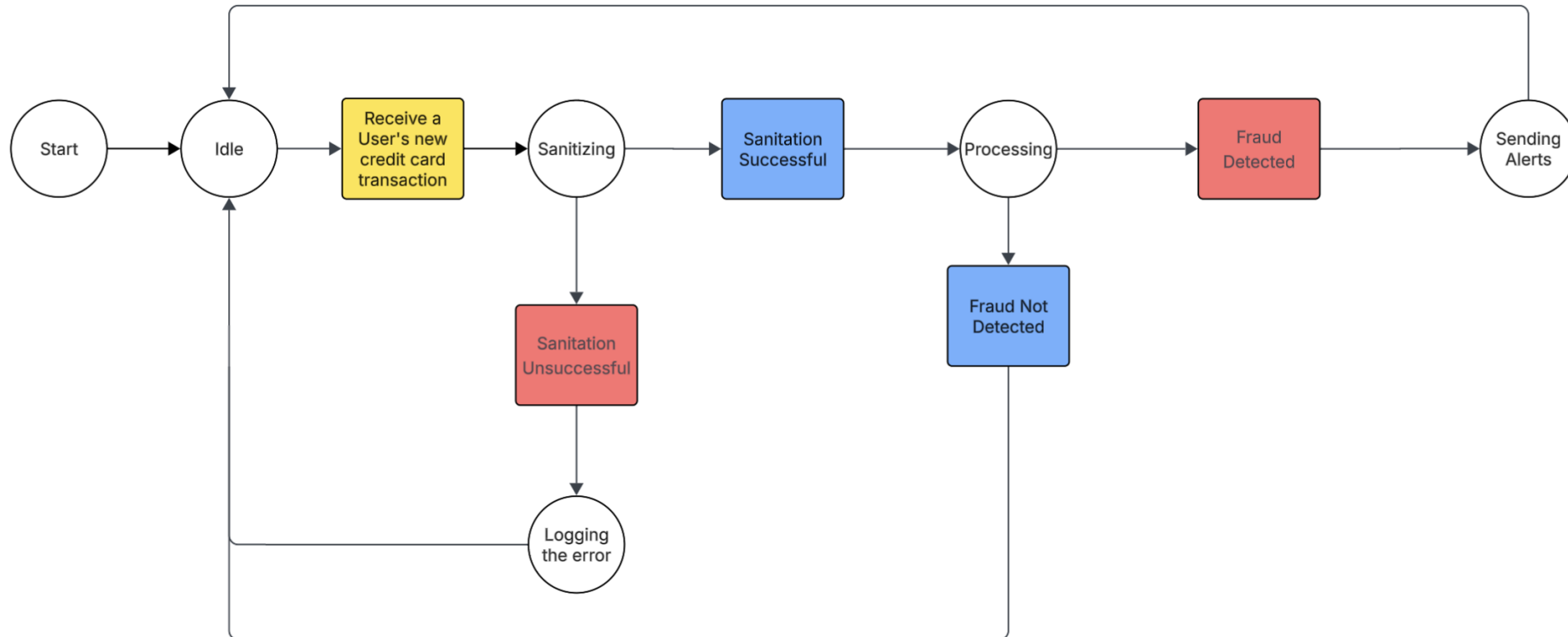
Selected for its strength with **tabular data** and internal mechanisms to effectively maximize **Recall** (catching fraud) and maintain high **Precision** (minimizing false positives).



Diagrams

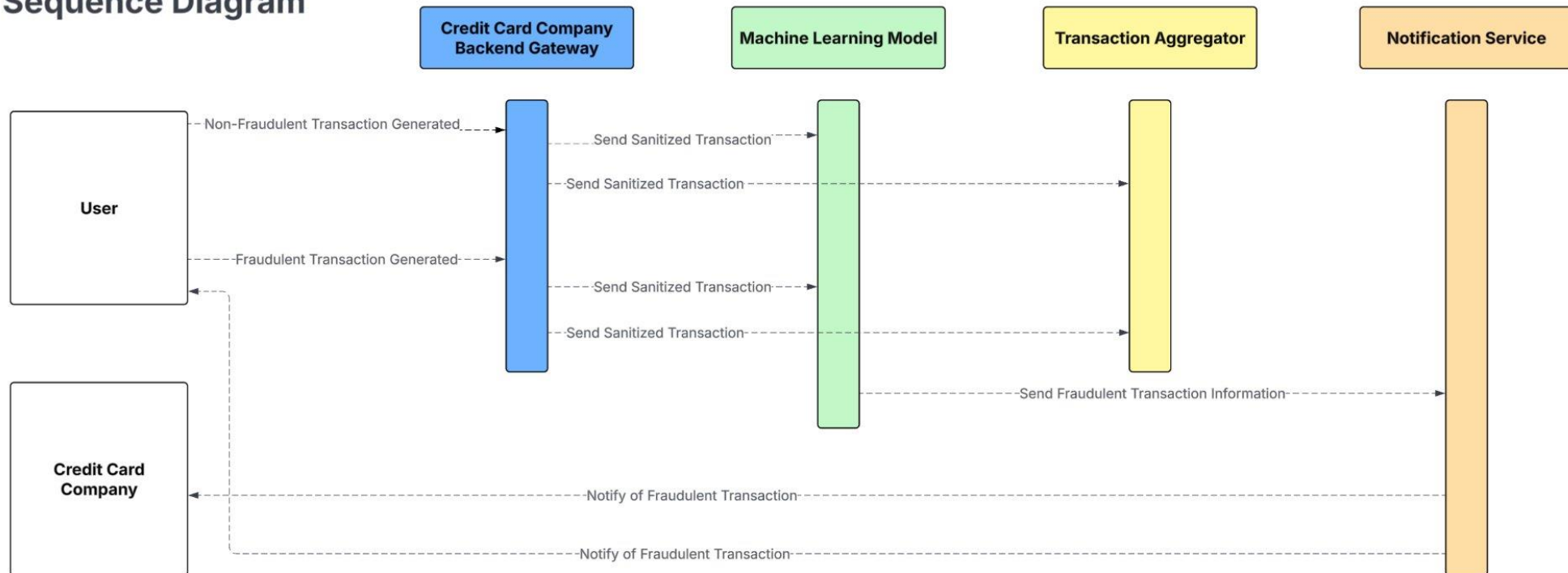
State Diagram

Credit Card Fraud Model State Diagram



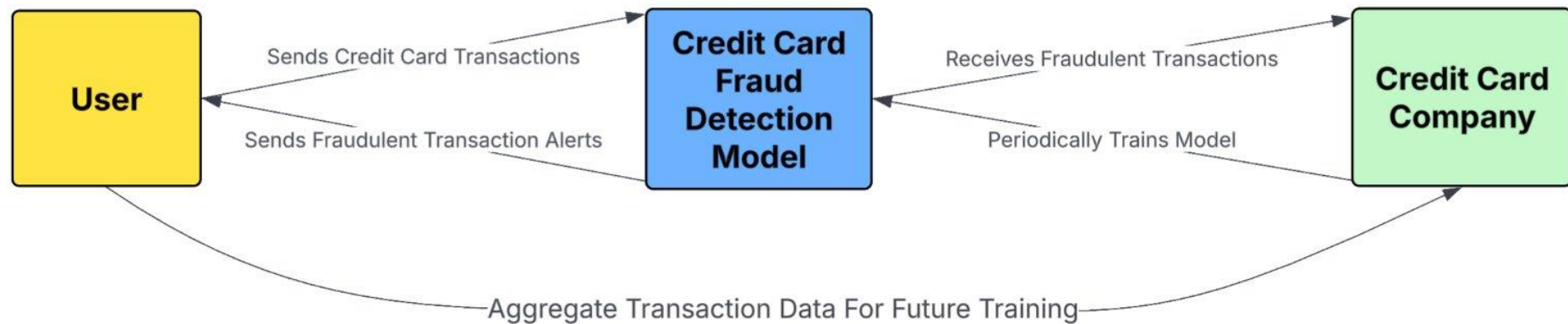
Sequence Diagram

**Credit Card Fraud Model
Sequence Diagram**



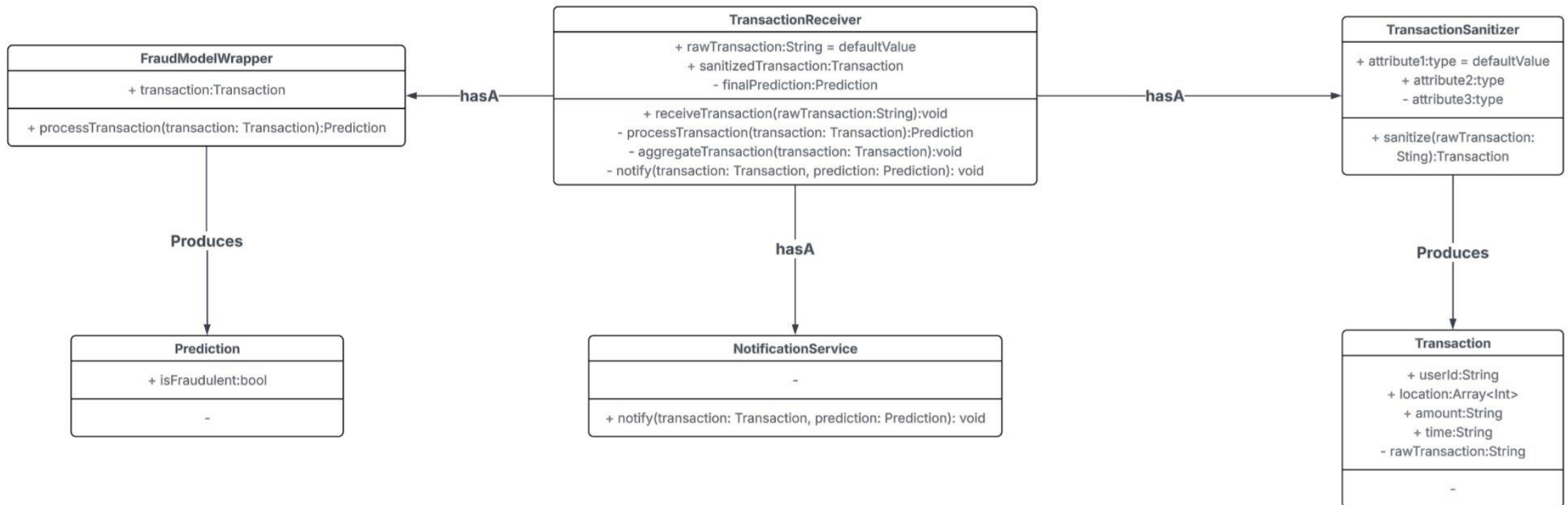
Context Diagram

Credit Card Fraud Model Context Diagram



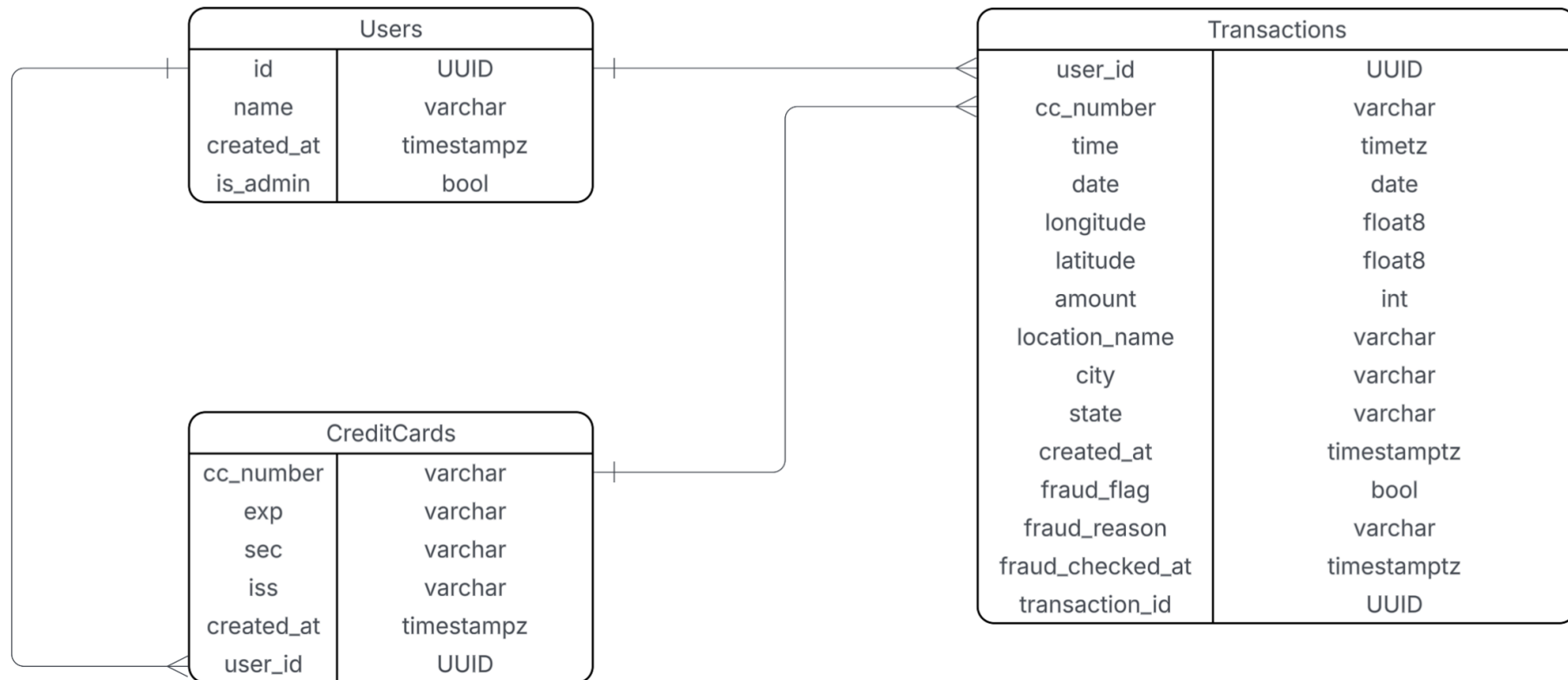
Class Diagram

Credit Card Fraud Model Class Diagram



Entity-Relationship Diagram

Credit Card Fraud Entity-Relationship Diagram



Sprint-1 Recap

- In our Sprint-1 we began the process of building our credit card fraud detection model
- We built a working model to detect credit card fraud that leveraged XGBoost, SMOTE, and credit card transaction data from Kaggle
- Our live demo consisted of a generator application that could generate a mock transaction with model features we could control
- To the right is a link to the start of our live demo to refresh your memory



Product Backlog

S.NO	User Story	Acceptance Criteria	Story Points
US_01	As a user, I want to securely register and log in so that my financial data remains protected.	User can sign up, log in, and authentication is validated using encrypted credentials and session tokens.	5
US_02	As a user, I want to link my credit card so that the system can analyze transactions for fraud.	User can add, validate, and save encrypted card details.	5
US_03	As the system, I want to ingest real-time transactions so that fraud detection happens instantly.	System successfully receives, validates, and stores transaction streams in the database.	8
US_04	As a fraud analyst, I want to review risky transactions so that I can quickly take action.	Analysts can view flagged transactions with risk scores and transaction details.	5
US_05	As a user, I want to receive alerts when suspicious activity occurs so that I can respond immediately.	Alerts trigger when fraud score exceeds threshold via dashboard notification.	5

Product Backlog

S.NO	User Story	Acceptance Criteria	Story Points
US_06	As the system, I want to run ML models to detect fraudulent patterns so that high-risk transactions are identified.	ML model generates fraud probability score for every transaction.	8
US_07	As an admin, I need to secure access to my dashboard so that no unauthorized person can view sensitive data.	Admin dashboard is protected with role-based access control and audit logging.	5
US_08	As a fraud analyst, I want a rule-based engine so I can define custom fraud rules.	Analysts can create and activate rules such as amount limits or location mismatch.	5
US_09	As a user, I want to view spending analytics so I can understand my financial behavior.	Dashboard displays charts for spending patterns, merchants, and time-based trends.	3
US_10	As a fraud analyst, I want to mark transactions as fraud or safe for model learning.	Analyst can update labels, add notes, and change case status.	5

Product Backlog

S.NO	User Story	Acceptance Criteria	Story Points
US_11	As a user, I want to file a chargeback for confirmed fraudulent transactions.	User can generate a chargeback request with required transaction information.	5
US_12	As an admin, I want to manage and verify financial institutions integrating with our system.	Admin can approve or reject banks after verifying API credentials.	5
US_13	As the system, I want to log all fraud-related events so that investigators can review them later.	Every fraud-related action is saved with timestamp, reviewer notes, and device metadata.	3
US_14	As a fraud analyst, I want heatmaps of unusual behavior so that anomalies are easier to detect visually.	System generates heatmaps for time-based, location-based, and merchant-based anomalies.	8
US_15	As a user, I want OTP verification for high-risk login attempts so that my account stays protected.	OTP is sent on suspicious logins and must be verified to continue.	5

Sprint-2 Stories

S.NO	User Story	Acceptance Criteria	Feature	Story Points
US_02	As a user, I want to link my credit card so that the system can start monitoring transactions for fraud.	User can add a card, Luhn validation passes, and card is stored securely.	Credit Card Linking	3
US_03	As the system, I want to ingest transactions so that fraud analysis can begin immediately.	System receives, validates, and stores transaction data automatically.	Transaction Ingestion	5
US_15	As an admin, I want analytic reports so that I can review overall fraud trends monthly.	Monthly reports generated automatically with charts.	Fraud Reporting Engine	5

Sprint-2 Stories

S.NO	User Story	Acceptance Criteria	Feature	Story Points
US_06	As the system, I want to run a machine learning fraud detection model so that fraudulent patterns can be identified.	Model generates probability score (0–1) for each transaction.	ML Fraud Detection Model	8
US_07	As a fraud analyst, I want to mark transactions as fraud or safe so that the model improves over time.	Analyst can label transactions and add case notes.	Case Management	3
US_10	As an admin, I want to manage system roles so that access controls are enforced.	Admin can assign roles (user, analyst, admin) with RBAC.	Role-Based Access Control	3

Sprint-2 Stories

S.NO	User Story	Acceptance Criteria	Feature	Story Points
US_12	As a user, I want to file a dispute for fraudulent charges so the bank can take action.	User fills dispute form linked to a specific transaction.	Fraud Dispute Filing	3
US_13	As the system, I want to maintain audit logs so that all actions related to fraud can be reviewed.	Every important action logged with timestamp, user, device info.	Audit Log System	2
US_14	As a user, I want to enable/disable fraud notifications so I have control over alerts.	User can toggle notification preferences in settings.	Notification Settings	2

Sprint-2 Test Cases

US_ID	TC_ID	Test Case	Steps	Execution Status	Result
US_05	TC_31	Trigger Fraud Alert for Suspicious Transaction	1. Login as Customer 2. Perform high-value transaction 3. Check system fraud alert 4. Validate alert is logged	Fraud alert should trigger	Pass
US_06	TC_33	Approve/Decline Suspicious Transaction	1. Login as Analyst 2. Open fraud alert 3. Approve/Decline 4. Add notes 5. Submit	Decision recorded	Pass
US_12	TC_39	Export Monthly Fraud Report	1. Login as Admin 2. Go to Reports 3. Generate Monthly Report 4. Download PDF/Excel	Report exported	Pass

Sprint-2 Test Cases

US_ID	TC_ID	Test Case	Steps	Execution Status	Result
US_09	TC_36	Customer Receives Fraud Notification	1. Trigger fraud 2. System blocks card 3. Notification sent SMS/Email 4. Verify message	Notification received	Pass
US_10	TC_37	False Positive Check – Legit Transaction	1. Perform normal transaction 2. Model evaluates 3. No fraud alert 4. Transaction succeeds	Transaction allowed	Pass
US_11	TC_38	Fraud Analyst Cannot View Restricted Cases	1. Login as Analyst A 2. Try accessing Analyst B's case 3. Access should be blocked	Unauthorized access blocked	Fail

Stories Completed And Not Completed

S.NO	User Story	Story Points	Completed	Date Completed
US_02	As a user, I want to link my credit card so that the system can start monitoring transactions for fraud.	3	3	Nov 4
US_03	As the system, I want to ingest transactions so that fraud analysis can begin immediately.	5	5	Nov 12
US_15	As an admin, I want analytic reports so that I can review overall fraud trends monthly.	5	0	-
US_06	As the system, I want to run a machine learning fraud detection model so that fraudulent patterns can be identified.	8	8	Nov 16
US_07	As a fraud analyst, I want to mark transactions as fraud or safe so that the model improves over time.	3	3	Nov 12
US_10	As an admin, I want to manage system roles so that access controls are enforced.	3	0	—

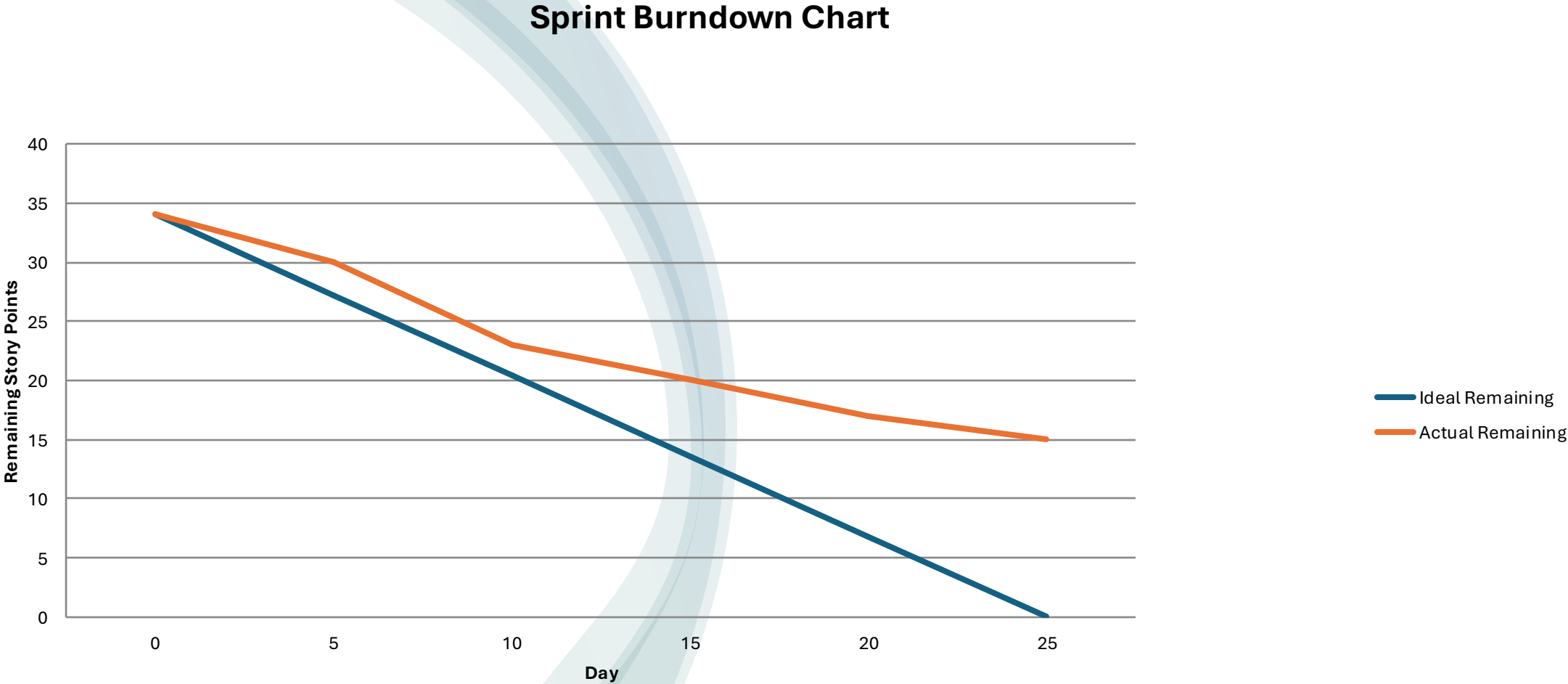
Stories Completed And Not Completed

S.NO	User Story	Story Points	Completed	Date Completed
US_12	As a user, I want to file a dispute for fraudulent charges so the bank can take action.	3	0	-
US_13	As the system, I want to maintain audit logs so that all actions related to fraud can be reviewed.	2	0	-
US_14	As a user, I want to enable/disable fraud notifications so I have control over alerts.	2	0	-

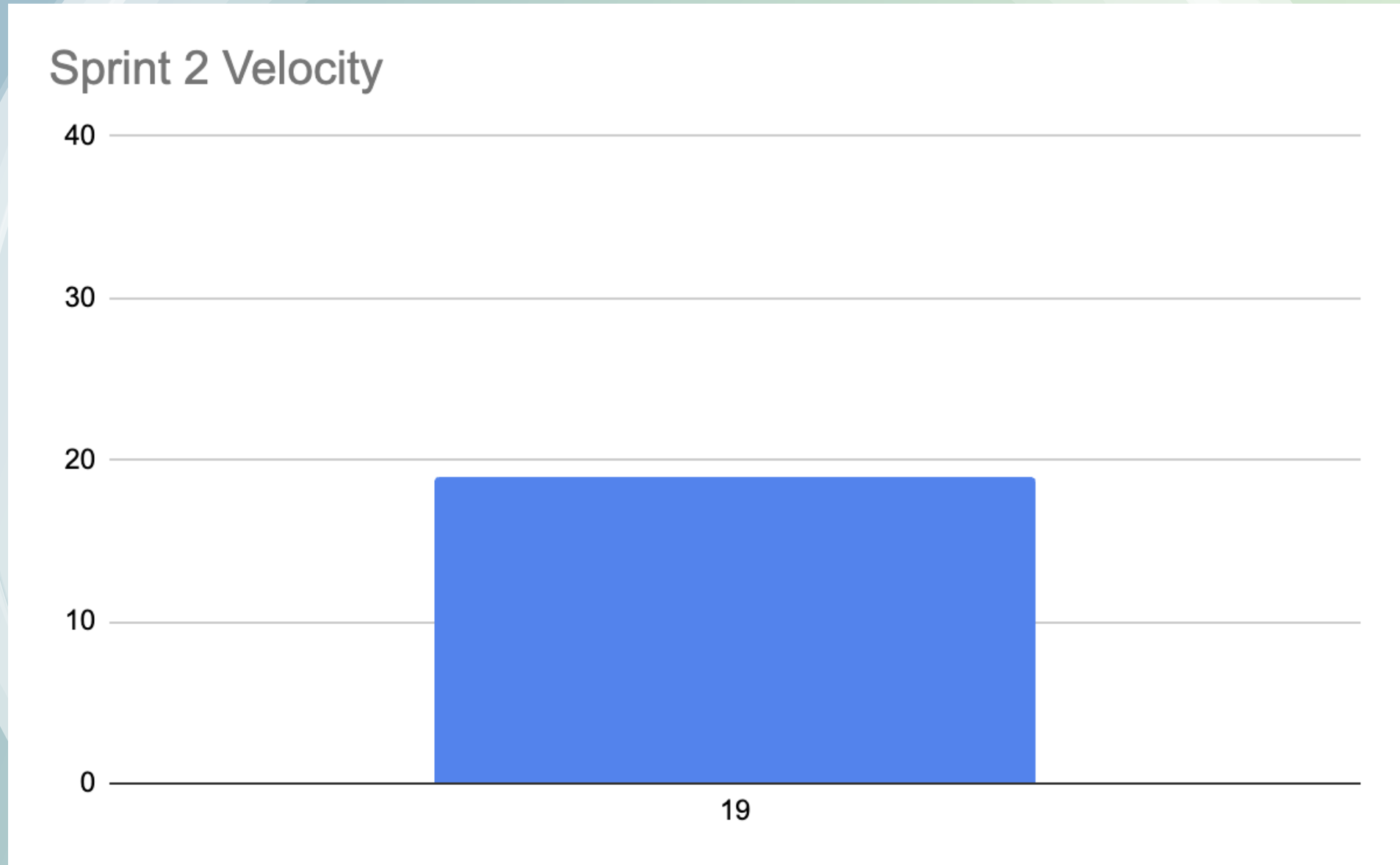


Metrics

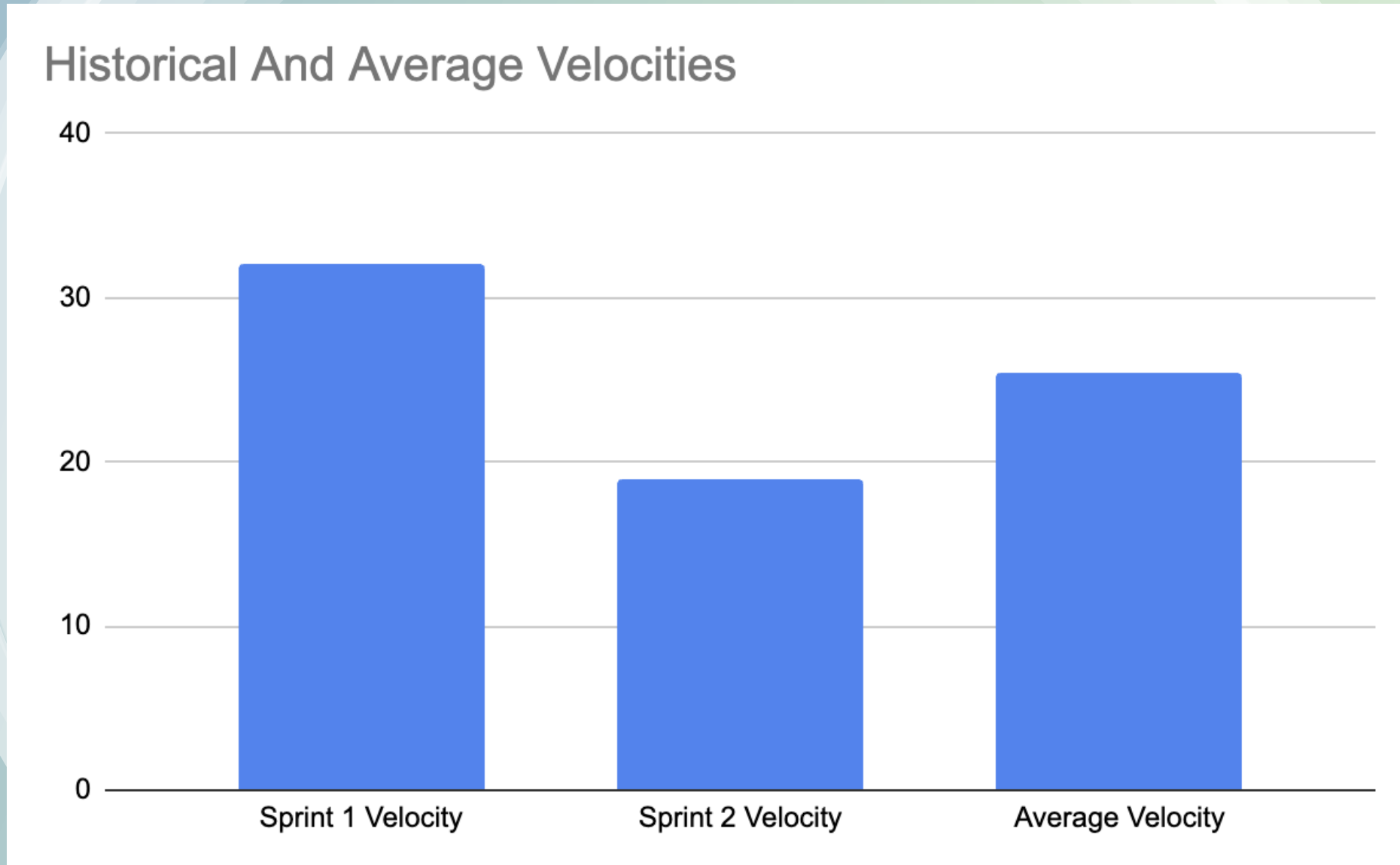
Burn-down chart



Sprint 2 Velocity Chart

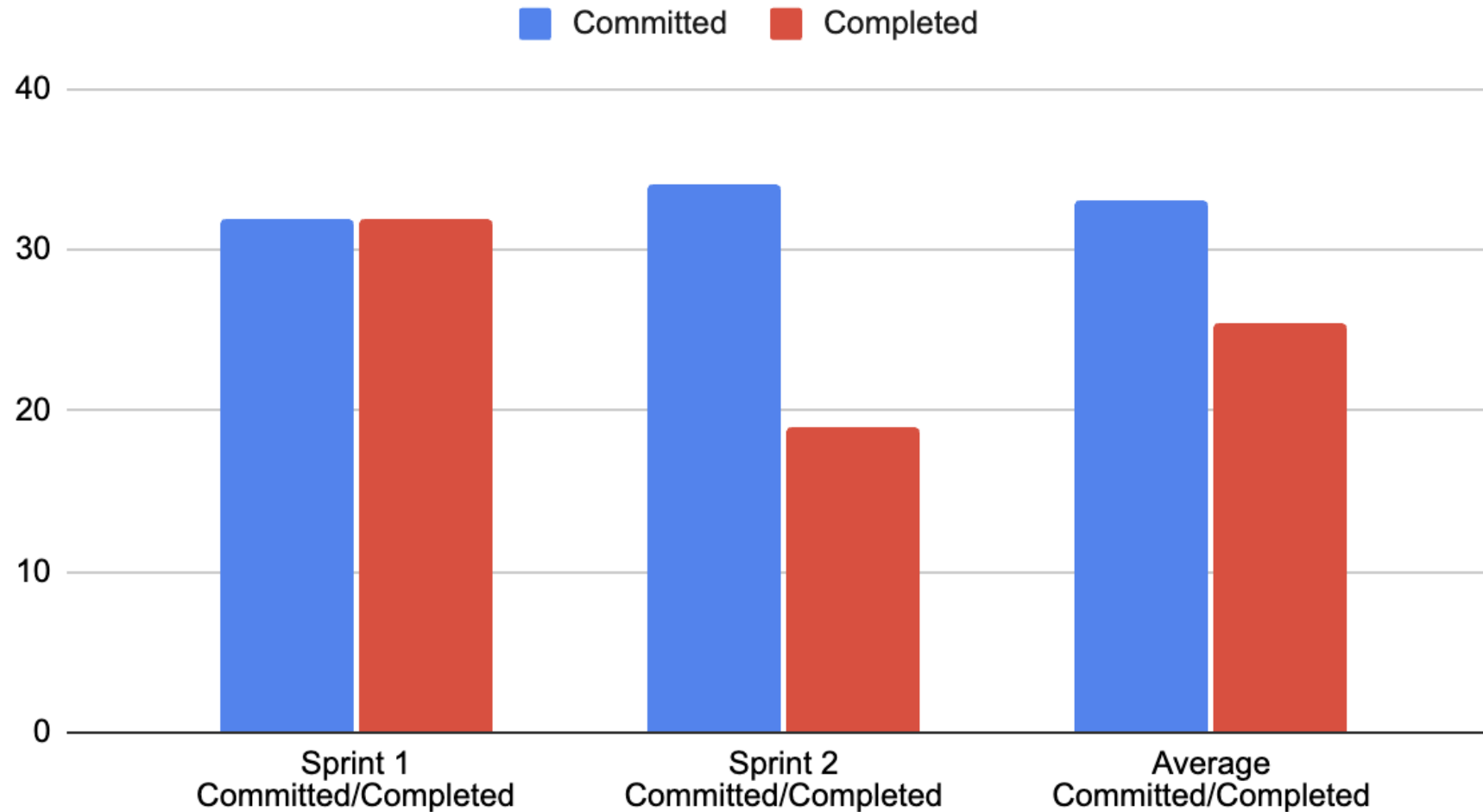


Historical & Average Velocities



Committed & Completed Ratios

Committed/Completed Ratios



Sprint Retrospective



- Video can be viewed here:
<https://youtu.be/Yizxb-Xj3eM>

Sprint Retrospective

IT Architects Sprint 2 Retrospective

IT Architects Sprint 2 Retrospective

What We Did Well	Votes
Met more consistently and spent time working together on front end and back end.	10
We have a functional backend with Supabase, it gave us postgres + edge functions	4
we actually did good scrum meetings and portray our thoughts well...	3
Came up with good plans and mockups for the app that will use our model	3
We understood the project better compared to Sprint 1.	3
Learning and working with new technologies.	3
Good pair programming working on features	2
Dependencies were managed better, and fewer last-minute issues happened.	2
Realizing that the project scope changed and adapted to the change while still incorporating the previous scope.	2
Worked at a more steady pace this sprint compared to last sprint	0
Collaborated more efficiently and used our time better.	0
Everyone have worked though project compare to last sprint	0
we did cover most of story point in this sprint..	0
Everyone was more active in our group chat and meetings.	0
we organized our work better than last sprint..	0
handled timings this time is better than last time	0
Flexibility and being honest in what we do well in and what we don't do well in.	0
Even though we didn't complete all of them our stories were better organized	0
Not feeling rushed.	0
had a great work when coming together for work, outcomes are much better and problems solved quickly	0
What We Can Do Differently	Votes

We should meet more frequently at a consistent time and stick to a strict meeting schedule.	5
We will define a realistic MVP so we don't overpromise and underdeliver.	4
we realize our project scope is way more than we expected so we should had decided our woking area earlier.	4
We need to be more talkative during meetings by sharing ideas, strengths/weaknesses, likes/dislikes about the project.	2
we should start reviewing our problems earlier..	2
we should start taking thought on each team member work while he is working on that particularly rather than after..	1
We started the tech paper far later than we should have	1
We should work on the power point presentation incrementally instead of waiting until the end of the sprint.	1
We need to be more proactive about our responsibilities and holding each other accountable for the work we do.	1
If someone is stuck, they will inform the team earlier instead of waiting until deadline.	1
We should follow what we have as our MVP as a guideline so we don't overextend our projects scope.	0
We need to restrict the scope of what we want our app to be	0
we should prioritize tasks according to its story points...	0
Although we improved from last sprint, we should work together even more often	0
we will begin working on diagrams, user stories, and backlog items from the first week.	0
In the mid of the sprint we should review our progress and tasks if needed.	0

Action Items	Votes
Need to meet everyday for at least an hour. (everyone)	0
Incorporate meeting notes. (Teams AI note taker)	0
Track sprint velocity (Tyler)	0

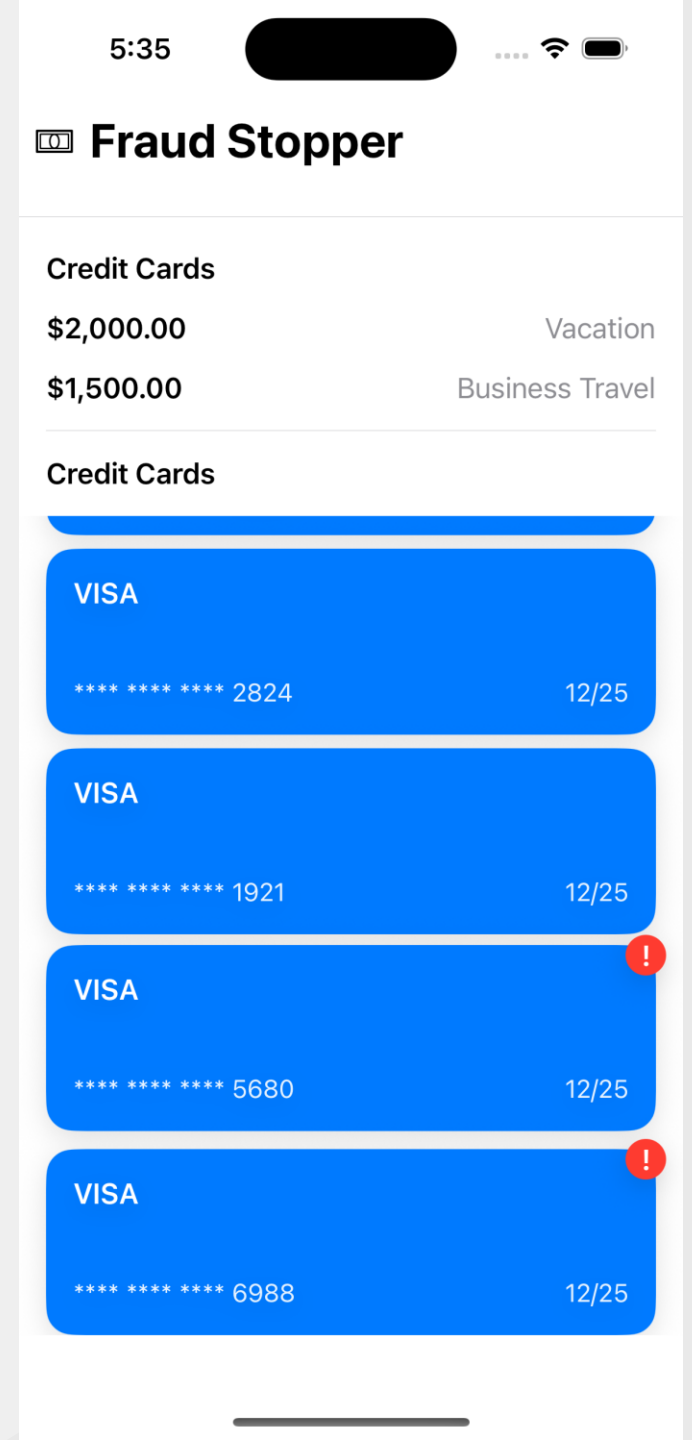
Manual Creation started by day 1 of Sprint 3 (Zenil/Xavier)	0
Tech Paper Draft 2 (Arth/Tyler)	0

Sprint-3 Planning Stories Carried Over

S.NO	User Story	Acceptance Criteria	Feature	Story Points
US_10	As an admin, I want to manage system roles so that access controls are enforced.	Admin can assign roles (user, analyst, admin) with RBAC.	Role-Based Access Control	3
US_12	As a user, I want to file a dispute for fraudulent charges so the bank can take action.	User fills dispute form linked to a specific transaction.	Fraud Dispute Filing	3
US_13	As the system, I want to maintain audit logs so that all actions related to fraud can be reviewed.	Every important action logged with timestamp, user, device info.	Audit Log System	2
US_14	As a user, I want to enable/disable fraud notifications so I have control over alerts.	User can toggle notification preferences in settings.	Notification Settings	2
US_15	As an admin, I want analytic reports so that I can review overall fraud trends monthly.	Monthly reports generated automatically with charts.	Fraud Reporting Engine	5

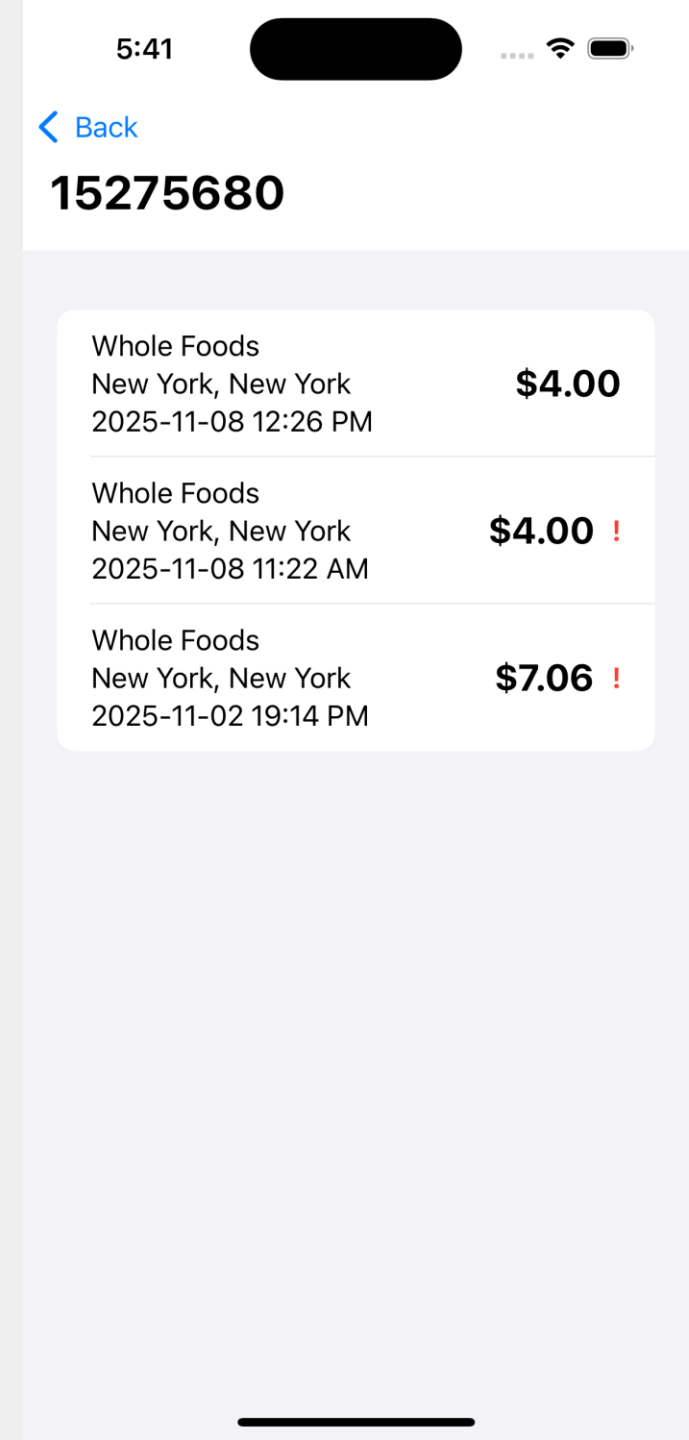
Application Screenshots

- The Homepage shows a list of user credit cards, cards that have experienced fraudulent transactions will have a red exclamation badge to indicate the fraud to the user
- The User can scroll their list of credit cards and can click into a credit card to view the transactions for that card (shown on the next page)



Application Screenshots

- This page shows a list of transactions for a specific credit card
- The credit card number is shown at the top of the list as a header
- A transaction row contains the location, city, state, and time of transaction along with the amount
- If a transaction is believed to be fraudulent it will have a red exclamation point to the right of its transaction amount



REST API

- All APIs routes require an API secret in the auth header in-order to access the route
- User ID information is also included in the http header rather than the URL

GET /functions/v1/app-middleware/user : Retrieves all user information such as name, email, and list of credit cards

POST /functions/v1/app-middleware/creditcard : Adds a new credit card to the user's account

```
{
  "userId": "...",
  "ccNumber": "...",
  "exp": "...",
  "sec": "...",
  "iss": "MASTERCARD | VISA | DISCOVER | etc."
}
```

GET /functions/v1/app-middleware/dailyreport?startTime=<...>&endTime=<...>

If the user is an admin they can use this route to access a daily report of fraudulent transactions

GET /functions/v1/app-middleware/transactions?ccNumber=<...>

Retrieves all transactions for a user, with the option to get transactions only for a specific credit card

REST API - Continued

POST /functions/v1/app-middleware/transactions : Create a new transaction

```
{
  "userId": "...",
  "ccNumber": "...",
  "longitude": 0.0,
  "latitude": 0.0,
  "amount": 400,
  "fraud_flag": false // false | true
}
```

PUT /functions/v1/app-middleware/transactions : Update the state of a transaction, primarily fraud status

```
{
  "fraud_flag": false // set to false | true
}
```

WIKI Page & Github Repository

[IT Architects Wiki Page](#)

[IT Architects Github Repository](#)

Project Demo

Thank You

